EDGE INTELLIGENCE · IOT FAULT DETECTION

Smart Motor Health Monitoring & Fault Detection System

Multi-sensor fusion and on-device decision making for real-time motor diagnostics — ESP32 · MPU6050 · ACS712 · Piezoelectric shock sensing.

HEALTH INDEX RANGE	RESPONSE LATENCY	CLOUD DEPENDENCY	UNIT COST
0–100%	<1 sec	NONE	$15

DEPT. OF ELECTRONICS & COMMUNICATION — VIT UNIVERSITY, VELLORE

PROJECT DOSSIER

OVERVIEW

What this system does

Electric motor failure — from overheating, mechanical wear, or sudden shock — accounts for over 80% of unplanned industrial downtime. This project puts diagnosis directly on the motor: an ESP32 microcontroller fuses four sensor streams into a single live **Health Index**, classifies the fault type locally, and cuts power automatically before damage occurs — no server round-trip required.

Core capabilities

- Real-time fusion of temperature, current, and dual vibration sensing
- Fully on-device (edge) fault classification — zero cloud latency
- Automatic relay-based safety shutdown on severe faults
- Instant Telegram alerts over Wi-Fi
- Local 16×2 LCD status readout

Why dual vibration sensing

The MPU6050 IMU catches broadband mechanical vibration (imbalance, misalignment). The piezoelectric disc catches sharp impulsive shocks the IMU misses. Together they discriminate fault types that single-sensor systems confuse.

ARCHITECTURE

System processing pipeline

Sensor Read → **Signal Conditioning** → **RMS Calculation** → **Health Index** → **Fault Classification** → **Relay + Alert + LCD**

$$H = 100 \times (w_T \cdot S_T + w_I \cdot S_I + w_V \cdot S_V + w_P \cdot S_P)$$

WEIGHTED HEALTH INDEX ACROSS TEMPERATURE, CURRENT, MPU6050 & PIEZOELECTRIC CHANNELS

HARDWARE

Sensor & component stack

COMPONENT	FUNCTION	INTERFACE
ESP32 Dev Module	Central processing, Wi-Fi, decision logic	—
NTC Thermistor (10kΩ)	Temperature / overheat detection	Analogue ADC
ACS712-05B	Current sensing / overload detection	Analogue ADC
MPU6050	6-axis vibration RMS (broadband)	I²C
Piezoelectric Disc	Impulsive shock detection	Analogue ADC
5V Relay Module	Automatic motor power cutoff	GPIO
16×2 LCD + I²C Backpack	Local status display	I²C

VALIDATION

Test results across fault conditions

CONDITION	TEMP (°C)	CURRENT (A)	MPU RMS	PIEZO ADC	HEALTH	STATUS
Normal	25	1.2	150	10	95%	NORMAL
Load	30	2.8	300	20	70%	MILD
Overload	34	5.0	600	38	40%	ABNORMAL
Critical (MPU)	31	4.7	1200	30	10%	SEVERE
Critical (Piezo)	27	3.8	350	900	15%	SEVERE

ECONOMICS

Bill of materials

COMPONENT	QTY	UNIT COST (INR)	TOTAL (INR)
ESP32 Dev Module	1	350	350
ACS712-05B Module	1	80	80
MPU6050 Module	1	70	70
NTC Thermistor 10k	1	10	10
Piezoelectric Disc	1	20	20
5V Relay Module	1	40	40
16×2 LCD + I²C	1	120	120
18650 Li-ion Cells	2	150	300
DC Gear Motor	1	120	120
Breadboard + Wires	1	100	100
Resistors / Misc.	—	—	50
TOTAL COST (SINGLE UNIT)			₹1,260 (~$15)

POSITIONING

Advantages over existing approaches

Edge Intelligence All fault decisions run locally on the ESP32 — no cloud latency, sub-second protective response.	**Dual Vibration Sensing** MPU6050 + piezoelectric combination discriminates fault types single-modality systems miss.
Low Cost Complete prototype under $15 — accessible for small-scale industrial and academic deployment.	**Automatic Safety Shutdown** Relay-based hardware interlock is fail-safe, independent of software state.

ROADMAP

Limitations & future scope

Current limitations

- ACS712 zero-current offset drifts with temperature
- Health index weights are empirically assigned, not ML-optimized
- Validated on a single small DC motor only
- Requires Wi-Fi; no GSM fallback yet

Planned enhancements

- On-device TinyML anomaly detection
- Custom PCB with EMI shielding
- MQTT / Grafana fleet dashboard
- Three-phase motor support (MCSA)